Higher National Diploma in Quantity Surveying

Second Year, 1st Semester Examination – 2018

QS 21023 – Building Economics 1

Instructions for candidates:
Answer Part A is compulsory
Answer 03 Questions from part-B.
Answer for each question must be started in a new page

No of pages : 04
No of questions: 06
Time: Three (03) hours

Part-A

1. The RIBA (Royal institute of British Architects) Plan of Work is the definitive framework for the contemporary building design and construction process. It provides a stage-by-stage model to assist all members of the project team to manage the process from inception to completion and beyond.

   1.1. Briefly describe about the RIBA plan of work and identify the key benefits of RIBA plan of work? (05 Marks)

   1.2. Feasibility studies are very important in appraisal (inception) stage of RIBA. Briefly explain the main feasibility studies are considered in Project appraisal stage. (05 Marks)

   1.3. In 2013 the RIBA announced an update to the RIBA plan of work. What are the new things included in RIBA plan of work 2013 and why changes were made in RIBA plan of work? (05 Marks)

   1.4. Identify and write down the major steps of the RIBA plan of work, from inception (appraisal) to completion of a building project. Who are the groups and professionals involved in each stage? (05 Marks)

   1.5. Quantity Surveyor plays a major role in RIBA Plan of work. Describe the Quantity surveyor's duties and responsibilities related to RIBA plan of work? (05 Marks)

(Total 25 Marks)

2. Write down short notes on the following:

a)	Construction industry by its activities	(03 Marks)
b)	Government influence on construction sector	(03 Marks)
c)	Direct stakeholders and indirect stakeholders	(03 Marks)
d)	"GDP" and "GNP" relevant to economics	(03 Marks)
e)	"Quality" , "Cost" and "Time" factors of a Project.	(03 Marks)

(Total 15 marks)

Part-B

3. Assume that you are the senior QS in a middle scale quantity surveying firm. You are requested to prepare a preliminary project estimate (Rough budget) for a proposed 50 room hotel project. Since the proposal is still in the feasible phase there is not a single sketch or drawing that has been expedited.

3.1. Explain why estimate should be more accurate and what are the factors affected for accuracy of estimate? (03 Marks)

3.2. What is the purpose of the preliminary estimating? Calculate the rough cost (preliminary) estimate using the data given. (08 marks)

Plinth area of Building	-	1,650 Square Meter (Sqm)
Plinth area rate	-	Rs 50,000 per Sqm
Architectural Design and Drawing	-	1.5% basic building cost
Water and Sanitation	-	10% basic building cost
AC, IT and Elec. and Mech.	-	14% of basic building cost
Other Cost	-	6% of basic Building Cost
Contingencies	-	5% of overall cost
Supervision charges	-	5% of overall cost

3.3. Explain factors affected for preparation of following Rates

a) Unit Rate of works item (03 Marks)

b) Hourly Rate of labour (03 Marks)

c) Hourly Rate of Machine (03 Marks)

(Total 20 Marks)

4. The quantity surveyor as a cost adviser should advise the design team with regard to design economics to educate the design team. The Quantity Surveyors are mainly divided into consultant's quantity surveyor and contractor's quantity surveyors

4.1. The function of the consultant quantity surveyor is broadly concerned with the control of the cost on construction projects. Briefly explain the duties and responsibilities of the consultant quantity surveyors? (05 Marks)

4.2. The contractor's Quantity Surveyor will be expected to explain the contractor why the loss situation has arisen and may even be expected to foresee potential loss-making positions developing and warn the contractor so that he may take steps that are necessary to minimize or reverse the loss.

a) Identify and explain such situations that lead to losses of the contractor? (03 Marks)

b) Identify the basic responsibility of contractor's Quantity Surveyor in a construction site? (03Marks)

4.3. Due to the environmental problems, an initially planned proposed multi storied 5-star hotel was changed to a single storied building. Since there is enough available land, the consultants have proposed an individual cottage type hotel village. This is basically changing a vertical construction to horizontal construction method. Discuss the problems that may cause on the cost of the project and cost control functions of the quantity surveyors practice? (05 Marks)

4.4. Explain Briefly "Present Value and Future Value", "Payback period or cost recovery period" and "Net Positive Value (NPV)" with subjected to the valuation process. (04 Marks)

(Total 20 Marks)

5. The construction industry is a wide industry that encompasses many professionals therefore in the project there is a building team. This building team can be subdivided into a design team, a contractor's team, a building material suppliers and regulators.

5.1. Find out minimum five design team members from the project and explain briefly their roles in construction industry? (05 Marks)

5.2. Identify and explain how the institutes and organizations are prepared to improve the quality of the construction works? (04 Marks)

5.3. Each construction project is unique and faces its own set of challenges and opportunities so identifying and managing risks can be tricky, but possible with careful planning and execution. Find out and briefly explain typical risk on a construction project? (05 Marks)

5.4. Identify and explain the main critical factors that result in construction project cost overrun? (03 Marks)

5.5. Briefly describe contributes of the construction industry to the economic development of a country? (03 Marks)

(Total 20 Marks)

6.

6.1. Demand is the representation of the various amounts of a product that consumers are willing and able to purchase at each of a series of possible prices during a specific period of time. Supply represents 'the amounts of a product that producers are willing and able to make available for sale at each of a series of prices during a specific period'

a) Briefly explain about demand curve and identify the reasons for downward sloping of the demand curve. (04 Marks)

b) Briefly explain about supply schedule and supply curve? (03 Marks)

c) Explain the equilibrium price by using supply and demand curve? (03 Marks)

6.2. Government adopts different types of policies at different cycle times due to upliftment of the construction industry in country. Identify and illustrate the policies to government to implement and uplift the construction industry? (04 Marks)

6.3. The Construction Industry Development Authority (CIDA) is an organization set up by the Government of Sri Lanka to develop and promote the domestic Construction Industry, Contractors, Professionals, Work Force, etc

a) Identify and explain the specialized field contractor s have to register under CIDA (03 Marks)

b) CIDA registration is a requirement for obtaining government contracts and there are over 2500 Construction Contractors registered with CIDA. Explain the purpose/benefits of registration scheme? (03 Marks)

(Total 20 Marks)